

Probability and Random Processes

Monsoon 2026

Exam: Mid Semester
Total Marks: 50

Date: 22 Sept 2026
Time: 11:00 AM to 12:30 PM

Instructions:

- All questions are compulsory.
- Clearly state any assumptions you make that are not specified in the question.
- No electronic devices other than calculators are allowed.

1. State whether the following statements are true or false. [10×1]

- F** a. $P(A \cap B) = P(A \cup B) - P(A) - P(B)$.
- T** b. For $A \subset \Omega$, $P(\Omega|A) = 1$.
- F** c. The posterior probability is always higher than the prior probability.
- F** d. If the events A and B are independent, then A and B conditioned on C are also independent.
- T** e. Geometric distribution models the number of attempts required for the first success in independent Bernoulli trials.
- F** f. $E[X + Y] = E[X] + E[Y]$ is true only if X and Y are independent. ~~$E(X+Y)$~~
- T** g. $\text{Var}(aX + b) = a^2 \text{Var}(X)$.
- T** h. $E[\frac{1}{X}] \neq \frac{1}{E[X]}$
- T** i. For $x_1 > 1$, $F_X(x_1) \leq F_X(x_1^2)$.
- F** j. Mean and Variance of the Poisson random variable are equal.

2. Answer the following questions in short. [9×2]

- State the axioms of probability.
- Prove the memorylessness property of exponential distribution.
- Find the expected value of $\mathbb{I}(A)$ such that $\mathbb{I}(A) = 1$ if event A occurs, otherwise $\mathbb{I}(A) = 0$.
- Consider that X is uniformly distributed over interval $[0, \theta]$. Find θ such that $E[X] = \text{Var}(X)$.
- Let X be a random variable that takes values from 0 to 9 with equal probabilities. Find the PMF of $Y = X \bmod 6$.
- Let X_1 and X_2 be identically and independently distributed random variables. Find the CDF of $X = \max(X_1, X_2)$.
- State the PMFs of Binomial and Geometric distributions.
- If $f_{X,Y}(x,y) = x/y$ such that $0 \leq X \leq 1$. Find the range of Y ?
- If $X_1 \sim \mathcal{N}(\mu_1, \sigma_1^2)$ and $X_2 \sim \mathcal{N}(\mu_2, \sigma_2^2)$, state the joint PDF of X_1 and X_2 .

3. Answer any three of the following questions. [3×4]

- Prove that $P(A \cup B|C) \leq P(A|C) + P(B|C)$.

$$P(A \cup B|C) = \frac{P((A \cup B) \cap C)}{P(C)}$$

$$P((A \cap C) \cup (B \cap C))$$

$$P(A \cap C) + P(B \cap C) - P(A \cap B \cap C)$$

$$P(C)$$

0 1 2 3 4 5 6 7 8 9

$$E[X|A_i] = \sum_x x P(X|A_i)$$

$$P(X) = \sum_y P(X|Y) P(Y)$$

$$\sum_x x P(X) \leq \sum_y \sum_x x P(X|Y) P(Y)$$

$$x = \sqrt{e^{-\lambda x}}$$

b. Show that

$$E[X] = \sum_i E[X|A_i] P(A_i),$$

such that $A_i \cap A_j = \emptyset$ for $i \neq j$.

c. Let $X \sim \text{Exp}(\lambda)$ and

$$Z = \begin{cases} X, & \text{with probability } p, \\ -X, & \text{with probability } 1-p. \end{cases}$$

Find the distribution of Z .

d. Show that

$$\text{Var}(X) = E[\text{Var}(X|Y)] + \text{Var}(E[X|Y]).$$

4. Answer any one of the following questions.

[1×10]

a. Consider N independent Bernoulli trials such that

$$X_n = \begin{cases} 1, & \text{with probability } p \\ 0, & \text{with probability } q = 1-p \end{cases}$$

for $n = 1, \dots, N$. Let a *run* denote maximal a block of consecutive identical outcomes and let R denote the number of runs in N -length sequence. Find the mean and variance of R .

b. Sita tosses a biased coin repeatedly. She wins if S consecutive head occurs before R consecutive tails, otherwise she loses. Find the probability that Sita wins.

ALL THE BEST!

$$E[(X^2) - E(X)^2] = E[E(X^2|Y) - (E(X|Y))^2] + E[E(X|Y)^2] - (E[E(X|Y)])^2$$

$$\begin{array}{r} 10110001 \\ \hline \end{array}$$

$$E_y(E[X_y^2] - E[X_y]^2) p^S + E[X^2] - E_y[E^2(X_y)] (1-p)p^S + (1-p)^2 p^S$$

$$E_y(E[X_y^2]) - E_y(E[X_y])^2 (1-p)^{R-1} p^S$$

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$$\sum P(Y) E_x[X^2 - E(X)^2] + E_y(E[X_y^2]) - E_y(E[X_y])^2$$

$$+ E_y(E[X_y^2]) - E_y(E[X_y])^2$$

$$E_y(E[X|Y] - E_y E[X|Y])$$